



if / if-else in Scratch — Take-Home Practice

Expert answer key — model responses for all tiers and rubric guidance.

SCRATCH

IF / IF-ELSE

3 TIERS

AGES 7-13

AI DECISION TREES

✓ ANSWER SHEET

NAME

ANSWER KEY — All Sections

DATE

Jul __, 2026

SECTION

Section 1 / 2 / 3

📌 **INSTRUCTOR NOTE:** Part 1 planning entries are student-specific — grade on completeness, not exact content. Bolded text = key concept for full credit. Accept equivalent wording throughout.

📌 PART 1 — PLAN YOUR PROJECT (ALL TIERS)

Think Before You Build

📌 FILL IN YOUR PLAN — GRADING GUIDE

My sprite's name / what it looks like:

Any specific description (e.g. "Whiskers, an orange cat")

Deduct 1pt if blank. Do not require specific sprite.

The condition I will check (IF __):

Must be a testable Scratch condition

e.g. <key [space] pressed?> or <touching edge?> — not vague like "something"

What happens when TRUE (THEN):

Concrete action (say, move, play sound)

Deduct if vague like "something happens".

What happens when FALSE (ELSE):

Different action from THEN branch

Shows student understands bifurcation — different from THEN.

📌 SKETCH YOUR SPRITE & STAGE

SCORING:

2 pts — sprite AND stage drawn, condition location marked

1 pt — any sketch attempted (partial)

0 pts — blank

Do not penalize artistic quality — award points for attempt and completeness of labeling.

🔍 QUICK CONCEPT CHECK — CORRECT ANSWERS

If condition is FALSE, the **if then** block...

✓ **skips its code** — because the **if** block only executes when the Boolean condition evaluates to TRUE. When FALSE, it does nothing.

Why is **if-else** always useful?

✓ **Because it always runs something** — it handles BOTH outcomes (true and false), so the program never ignores a situation. There is always a response.

📌 PART 2 — BUILD IT! TIER ANSWERS & RUBRIC

Open Scratch and Follow Your Tier's Steps

☆ TIER A — STARTER (4 pts)

TASK: ONE IF-ELSE BLOCK — KEY PRESS CHANGES SPRITE'S WORDS.

- Project opened; sprite clicked.
- **forever** loop + **if-else** block added inside.
- **<key [space] pressed?>** in the diamond condition.
- THEN & ELSE both customized (not default words).
- "My sprite said" section filled in for both cases.

✓ **Full credit: both THEN/ELSE filled + logged. –1pt if either unchanged from default.**

☆ **Bonus: +1pt if 2nd key condition added.**

☆☆ TIER B — EXPLORER (4 pts)

TASK: ONE PLAIN [IF] + ONE [IF-ELSE], BOTH IN A [FOREVER] LOOP.

- **if** **<touching edge?>** → say "Oops! Wall!" (plain if, no else).
- **if-else** **<UP arrow?>** → THEN move 10 / ELSE different action.
- Both blocks inside **forever** loop. same
- All 4 test situations logged (✓ or X).

✓ **"Why if-else?" model answer: "Because if-else guarantees the sprite does something in both cases — with just [if], pressing nothing gives no response."**

☆☆☆ TIER C — BUILDER (4 pts)

TASK: 3 CONDITIONS INCL. NESTED + AI REFLECTION.

- **if-else** touching edge → two distinct responses.
- **if-else** key press → move THEN / wait ELSE.
- Nested **if** INSIDE edge THEN: space → "I escaped!" / "I'm stuck!"

NESTED LOGIC MAP — CORRECT ANSWER:

```
IF touching edge →
  THEN: IF space pressed →
    THEN: "I escaped!"
    ELSE: "I'm stuck!"
  ELSE: [normal walk response]
```

✓ **AI connection must link nested ifs to classifier: "A nested conditional is like an AI image classifier because both check one feature first, then check a second feature inside to reach a final decision." –2pts if AI connection absent.**

🗣️ PART 3 — REFLECTION (ALL TIERS)

Look Back at What You Built

📄 ANSWER ALL 3 QUESTIONS — MODEL ANSWERS

1. What did your **if-else** block do in your project?

✓ Model: "My if-else block made the sprite say 'Woo!' when I pressed space, and 'Waiting...' when I didn't — so it always responded."

Key: student must explain BOTH branches.

2. What was the hardest part? How did you figure it out?

✓ Full credit requires BOTH: (a) **naming something specific** that was hard, AND (b) **describing how they resolved it** — not just "I asked for help."

3. One question for class tomorrow:

✓ Any sincere conceptual question earns full credit. E.g. "Can you nest more than 2 ifs?" or "What if both conditions are true?" Deduct 1pt if left blank.

📊 SCORING RUBRIC

SECTION	PTS
Planning (plan + sketch attempted)	2
Tier A — one working if-else ; words filled; tested	4
Tier B — two conditions; test table done; explanation written	4
Tier C — 3 conditions incl. nested; logic map + AI reflection	4
Reflection — all 3 questions answered thoughtfully	2
☆ Bonus — extra condition beyond requirement	+1
TOTAL	12+1

📈 FEELING SCALE — TEACHER GUIDE

- 🙄 **Lost** — needs re-teaching
- 😞 **Getting there** — needs practice
- 😊 **I get it!** — on track
- 🚀 **Teach it!** — extend/accelerate

